

Data Structures

Lab task 8

Convert Infix → Postfix

Infix:

$$1. A + B * (C \wedge D - E) \wedge (F + G * H) - I$$

STEP	SCAN	OUTPUT
A		A
+	+	A
B	+	AB
*	+	AB
(	+	AB
C	+	ABC
^	+	ABC
D	+	ABCD
-	+	ABCD^
E	+	ABCD^E
)	+	ABCD^E-
^	+	ABCD^E-
(	+	ABCD^E-
F	+	ABCD^E-F
+	+	ABCD^E-F
G	+	ABCD^E-FG
*	+	ABCD^E-FG
H	+	ABCD^E-FGH
)	+	ABCD^E-FGH*
-	-	ABCD^E-FGH*+^*+
I	-	ABCD^E-FGH*+^*+I
End		ABCD^E-FGH*+^*+I-

$$2- A + (B * C - (D / E \wedge F) * G) * H$$

STEP	SCAN	OUTPUT
A		A
+	+	A
(	+	A
B	+	AB
*	+	AB
C	+	ABC
-	+	ABC*
(	+	ABC*
D	+	ABC*D
/	+	ABC*D
E	+	ABC*DE
^	+	ABC*DE
F	+	ABC*DEF
)	+	ABC*DEF^/
*	+	ABC*DEF^/

G	+ (- *	ABC*DEF^/G
)	+	ABCDEF^/G-
*	+ *	ABCDEF^/G-
H	+ *	ABCDEF^/G-H
End		ABCDEF^/G-H*+

Convert Infix → Prefix

1. $A + (B * (C + (D * E)))$

Reverse:

$((E * D) + C) * B + A$

STEP	SCAN	OUTPUT
(	(	
(	((	
E	((	E
*	((*	E
D	((*	ED
)	(	ED*
+	(+	ED*
C	(+	ED*C
)		ED*C+
*	*	ED*C+
B	*	ED*C+B
+	+	EDC+B
A	+	EDC+BA
End		EDC+BA+

Reverse Output:

$+ A * B + C * D E$

2. $A * (B + C * (D - E)) + F / G$

Reverse:

$G / F + ((E - D) * C + B) * A$

STEP	SCAN	OUTPUT
G		G
/	/	G
F	/	GF
+	+	GF/
(	+	GF/
(	+	GF/
E	+	GF/E
-	+	GF/E
D	+	GF/ED
)	+	GF/ED-
*	+	GF/ED-

C	+ (*	GF/ED-C
+	+ (+	GF/ED-C*
B	+ (+	GF/ED-C*B
)	+	GF/ED-C*B+
*	+ *	GF/ED-C*B+
A	+ *	GF/ED-C*B+A
End		GF/ED-CB+A+

Reverse

+ * A + B * C - D E / F G

Evaluate the following expressions:

- 56+2*34+-

STEP	OUTPUT
5	5
6	5, 6
+	11 (5+6)
2	11, 2
*	22 (11x2)
3	22, 3
4	22, 3, 4
+	22 , 7 (3+4)
-	15 (22-7)

- - * 4 5 + 2 3

STEP	OUTPUT
3	3
2	2, 3
+	5 (2+3)
5	5 , 5
4	4, 5, 5
*	20 (4x5) , 5
-	15 (20-5)

- + * 2 3 4

STEP	OUTPUT
4	4
3	3, 4
2	2, 3, 4
*	6 (2x3) , 4
+	10 (6+4)

● 93*42/-

STEP	OUTPUT
9	9
3	9, 3
*	27 (9x3)
4	27, 4
2	27, 4, 2
/	27, 2 (4/2)
-	25 (27-2)